Huawei Access Node WHI-ACC-50 SOP for QoS / Congestion Event (Microbursts, interface tail drops, QoS queue starvation) - Whitefield East POP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5947

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a QoS / Congestion Event failure on the Huawei Access Node WHI-ACC-50 at the Whitefield East POP. The failure is characterized by the ALARM_5947 alarm code and is categorized as Critical in severity. The event is likely caused by microbursts, interface tail drops, or QoS queue starvation on the access node. This equipment failure domain is specific to Huawei Access Nodes and occurs when the network is subjected to bursty traffic patterns that exceed the node's buffer capacity, leading to packet drops and subsequent QoS degradation. This is due to the hardware-based QoS mechanisms implemented on the node, which are designed to mitigate congestion by prioritizing traffic based on class of service (CoS) and rate limiting. However, when the buffer is overwhelmed by a sudden influx of packets, the node's QoS mechanisms are unable to keep up, resulting in packet drops and subsequent QoS degradation.

Pre-requisite Checks

- check: Verify that all redundant links at the Whitefield East POP are up and functioning correctly before proceeding with the diagnostic and resolution steps.

- check: Ensure that the remote hands team is dispatched to the Whitefield East POP to facilitate any necessary physical interventions.

Potential Root Causes

Cause	Probability
Hardware-based QoS mechanisms overwhelmed by bursty traffic	High
Misconfigured QoS settings on the access node	Medium
Software bugs or glitches affecting the QoS mechanisms	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics for each QoS class of service (CoS), Description: This command displays the current queue statistics for each QoS CoS on the access node, including the number of packets in each queue, the average queue depth, and the packet drop count. This information can be used to identify any QoS queue starvation or tail drops that may be contributing to the congestion event.
```

```
Command: display interface queue statistics, Expected Output:
Interface queue statistics, including packet drop count and average
queue depth, Description: This command displays the current interface
queue statistics, including the packet drop count and average queue
depth for each interface on the access node. This information can be
used to identify any interface tail drops or packet drops that may be
contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS settings, Command: configure qos queue settings, Description: Verify that the QoS settings on the access node are correctly configured and adjust them as necessary to prevent QoS queue starvation or tail drops. This may involve adjusting the packet buffer size, adjusting the QoS CoS priorities, or implementing additional QoS mechanisms such as policing or shaping.

Step 2: Step: Increase buffer capacity, Command: configure buffer settings, Description: Increase the buffer capacity on the access node to prevent QoS queue starvation or tail drops. This may involve increasing the packet buffer size or implementing additional buffer mechanisms such as buffer policing or shaping.

Step 3: Step: Disable hardware-based QoS mechanisms, Command: configure qos hardware settings, Description: Disable the hardware-based QoS mechanisms on the access node to prevent QoS queue starvation or tail drops. This may involve disabling the QoS hardware acceleration or implementing a software-based QoS solution.

Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue
statistics for each QoS class of service (CoS) show no signs of
congestion or QoS queue starvation, Description: This command is used
to verify that the QoS queue statistics show no signs of congestion or
QoS queue starvation after the resolution steps have been implemented.

Command: display interface queue statistics, Expected Output:
Interface queue statistics show no signs of interface tail drops or
packet drops, Description: This command is used to verify that the
interface queue statistics show no signs of interface tail drops or
packet drops after the resolution steps have been implemented.
```

Roll-back Procedures

- Step: Revert QoS settings, Command: undo configure qos queue settings, Description: Revert the QoS settings on the access node to their original values in case the resolution steps failed to resolve the issue.

- Step: Revert buffer capacity, Command: undo configure buffer settings, Description: Revert the buffer capacity on the access node to its original values in case the resolution steps failed to resolve the issue.

- Step: Re-enable hardware-based QoS mechanisms, Command: undo configure qos hardware settings, Description: Re-enable the hardware-based QoS mechanisms on the access node in case the resolution steps failed to resolve the issue.